

AI Marketing Campaign Intelligence

A step-by-step guide explaining what every file is, what you did in the project, and exactly how to upload and present it on GitHub and LinkedIn.

Project Story

In this project, I built an end-to-end marketing analytics solution that turns raw campaign data into business decisions. I generated a realistic campaign dataset, cleaned it with Python, analyzed it with SQL, calculated marketing KPIs, produced dashboard-ready outputs, and wrote executive recommendations.

Best GitHub repository name

ai-marketing-campaign-intelligence

One-line project description

End-to-end SQL and Python marketing analytics project that converts campaign data into executive recommendations.

What You Did

1. Defined a business problem: marketing teams need to know which campaigns deserve more budget.
2. Created a realistic synthetic dataset with campaign spend, impressions, clicks, leads, conversions, revenue, cities, channels, and audience segments.
3. Designed a SQL schema for campaign analytics.
4. Wrote SQL queries to answer business questions about channel performance, campaign ROI, city opportunity, and audience segmentation.
5. Built a Python pipeline to clean data, calculate KPIs, aggregate results, and create charts.

6. Created dashboard-ready CSV files for Power BI or Tableau.
7. Wrote an executive report with findings and recommendations.
8. Prepared a GitHub README, dashboard concept, notebook walkthrough, and LinkedIn post.

File-by-File Guide

FILE / FOLDER	WHAT IT IS	UPLOAD TO GITHUB?	HOW TO EXPLAIN IT
README.md	Main project page.	Yes	This explains the business problem, tools, workflow, findings, and how to run the project.
requirements.txt	Python package list.	Yes	This helps another person install the packages needed to run the analysis.
src/generate_synthetic_data.py	Data generation script.	Yes	This creates the realistic marketing campaign dataset used in the project.
src/run_analysis.py	Main analytics pipeline.	Yes	This cleans the data, calculates KPIs, creates processed files, builds SQLite tables, and generates charts.
sql/01_schema.sql	Database schema.	Yes	This shows I understand how to structure data for SQL analytics.
sql/02_business_questions.sql	Business analysis queries.	Yes	These SQL queries answer manager-level questions about ROAS, CPA, revenue, and market opportunity.
data/raw/marketing_campaign_data.csv	Raw dataset.	Yes	This is the starting dataset before analysis.
data/processed/*.csv	Cleaned and aggregated outputs.	Yes	These files are ready for dashboards in Power BI, Tableau, or Excel.

FILE / FOLDER	WHAT IT IS	UPLOAD TO GITHUB?	HOW TO EXPLAIN IT
outputs/charts/*.png	Generated charts.	Yes	These visuals summarize revenue, ROAS, weekly trends, top campaigns, and audience performance.
dashboard/dashboard.html	HTML dashboard concept.	Yes	This gives managers a visual preview of how the dashboard would look.
dashboard/dashboard_wireframe.md	Power BI/Tableau build plan.	Yes	This explains how to build the final BI dashboard pages.
notebooks/analysis_walkthrough.ipynb	Notebook walkthrough.	Yes	This lets recruiters quickly inspect the analytical thinking process.
docs/executive_report.md	Management report.	Yes	This converts the technical analysis into business recommendations.
docs/linkedin_post.md	LinkedIn post template.	Optional	Use this to announce the project professionally.
data/processed/marketing_campaigns.sqlite	Local SQLite database.	No	This is generated locally and is ignored by GitHub through <code>.gitignore</code> .

How to Upload to GitHub

Option 1: GitHub Website

1. Go to `github.com` and create a new repository.
2. Name it `ai-marketing-campaign-intelligence`.
3. Add this description: `End-to-end SQL and Python marketing analytics project that converts campaign data into executive recommendations.`
4. Keep it public.
5. Do not add a new README on GitHub because this folder already has one.
6. Upload all files and folders from `/Users/deepak/Documents/My Projects/Git Hub/Project 1`.
7. Commit with the message: `Initial end-to-end marketing analytics project`.

Option 2: Terminal Commands

Run these commands after creating the empty GitHub repo online:

```
cd "/Users/deepak/Documents/My Projects/Git Hub/Project 1"
git init
git add .
git commit -m "Initial end-to-end marketing analytics project"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/ai-marketing-campaign-intelligence
git push -u origin main
```

What to Show Recruiters

- Start with the README because it tells the full business story.
- Show `docs/executive_report.md` to prove business communication.
- Show `sql/02_business_questions.sql` to prove SQL ability.
- Show `src/run_analysis.py` to prove Python and analytics workflow.
- Show `outputs/charts/` and `dashboard/dashboard.html` to prove visualization and executive thinking.

How to Present This in an Interview

"I built this project to connect my marketing background with analytics. The business problem was campaign budget allocation. I created a realistic dataset, used SQL and Python to analyze campaign performance, calculated KPIs like ROAS and CPA, and converted the findings into recommendations a manager could act on."